12 Data collected in the twentieth century showed that the probability of a randomly selected person having blue eyes was 0.08. A medical researcher believes that the probability in 2024 is less than this so they decide to carry out a hypothesis test at the 5% significance level.

(a) Write down suitable hypotheses for the test, defining the parameter used. **[2]**

(b) Assuming that the probability that a person selected at random has blue eyes is still 0.08, calculate the probability that 3 or fewer people in a random sample of 92 people have blue eyes. **[1]**

The researcher collects a random sample of 92 people and finds that 3 of them have blue eyes.

(c) Use your answer to part **(b)** to carry out the test, giving your conclusion in context. **[3]**

12(a)

12(b)

12(c)